

Computer Science & IT

Discrete Mathematics

Mathematical Logic

Lecture No. 07

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Recap of Previous Lecture



Topic

Sentences with multiple quantifiers

Topic

Relationship diagram

Topics to be Covered



Topic

Rules of inferences with respect to predicate logic



Topic

Important logical implications and equivalences



Topic

Practice questions



H.W. Consider following two predicate formula

$$F_1: \quad \forall x \{ P(x) \leftrightarrow Q(x) \} \longrightarrow \{ \forall x \{ P(x) \} \leftrightarrow \forall x \{ Q(x) \} \}$$

$$F_2: \quad \{ \forall x \{ P(x) \} \leftrightarrow \forall x \{ Q(x) \} \} \longrightarrow \forall x \{ P(x) \leftrightarrow Q(x) \}$$

Which of the following is true

- (a) Only F_1 is valid
- (b) Only F_2 is valid
- (c) Both F_1 and F_2 are valid
- (d) Neither F_1 nor F_2 is valid

H.W. Consider following two predicate formula

$$F_1: \quad \forall x \{ P(x) \leftrightarrow Q(x) \} \longrightarrow \{ \forall x \{ P(x) \} \leftrightarrow \forall x \{ Q(x) \} \}$$

$$U = \{1, 2, 3\}$$

$$P(1) \leftrightarrow Q(1) = T$$

$$P(2) \leftrightarrow Q(2) = T$$

$$P(3) \leftrightarrow Q(3) = T$$

$$F \leftrightarrow F = T$$

$$P(1) \leftrightarrow Q(1) = T$$

$$P(2) \leftrightarrow Q(2) = T$$

$$P(3) \leftrightarrow Q(3) = T$$

$$P(1) \leftrightarrow Q(1) = T$$

$$P(2) \leftrightarrow Q(2) = T$$

$$P(3) \leftrightarrow Q(3) = T$$

$$F \leftrightarrow F = T$$

$$T \leftrightarrow T = T$$

- Whenever LHS is true, RHS will also be true so F_1 is valid
- When RHS is false, then LHS will also be false, i.e. main implication will be $(F \rightarrow F = T)$. $\therefore \rightarrow$ will never return false Hence F_1 is valid

H.W. Consider following two predicate formula

$$F_2: \{ \forall x \{ P(x) \leftrightarrow Q(x) \} \longrightarrow \forall x \{ P(x) \leftrightarrow Q(x) \}$$

$$\begin{array}{ccc} T & \longleftrightarrow & T \\ \uparrow & & \uparrow \\ \text{P will be true} & & \text{Q will be} \\ \text{for all } x & & \text{true for all } x \end{array} \equiv T$$

$\Rightarrow$ i.e., $\forall x (P(x) = T \text{ \& } Q(x) = T)$
 $\therefore \forall x \{ P(x) \leftrightarrow Q(x) \}$
will be true

$$\begin{array}{ccc} F & \longleftrightarrow & F \\ \uparrow & & \uparrow \\ \text{P is false} & & \text{Q is false} \\ \text{for some } x & & \text{for some } x \end{array} \equiv T$$

Let $U = \{1, 2, 3\}$ & $P(1) = F$ & $Q(1) = F$
 $P(2) = T$ & $Q(2) = F$

L.H.S. is true

R.H.S.

$$\{ P(1) \leftrightarrow Q(1) \} = T$$
$$\{ P(2) \leftrightarrow Q(2) \} = F$$

$\therefore \forall x \{ P(x) \leftrightarrow Q(x) \}$
will be false
i.e. R.H.S. = False

but

R.H.S. is false

Hence F_2 is invalid

H.W. Consider following two predicate formula

$$F_1: \quad \forall x \{ P(x) \leftrightarrow Q(x) \} \xrightarrow{\text{valid}} \{ \forall x \{ P(x) \} \leftrightarrow \forall x \{ Q(x) \} \}$$

$$F_2: \quad \{ \forall x \{ P(x) \} \leftrightarrow \forall x \{ Q(x) \} \} \xrightarrow{\text{invalid}} \forall x \{ P(x) \leftrightarrow Q(x) \}$$

Which of the following is true

- ☒ (a) Only F_1 is valid
- (b) Only F_2 is valid
- (c) Both F_1 and F_2 are valid
- (d) Neither F_1 nor F_2 is valid



Topic : Rules of Inferences w.r.t. predicate logic

- ❑ In predicate logic, we can use some additional inference rules, along with all the rules of inference we have discussed in propositional logic.

- ❑ Additional Rules of Inference w.r.t. Predicate Logic
 1. Universal Instantiation (Universal Specification)
 2. Universal Generalization
 3. Existential Instantiation (Existential Specification)
 4. Existential Generalization



Topic : Universal Instantiation

Universal specification



$$\forall x \{ P(x) \}$$

is valid Rule of inference
∴ $P(a)$ is true for all 'a' in the domain



Topic : Universal Generalization

$P(c)$ is true for all 'c' in the domain is valid

$$\therefore \forall x (P(x))$$

We can use any variable

} i.e. $\forall z \{ P(z) \}$ is also a }
Valid inference }



Topic : Existential Instantiation



$$\exists x \{ P(x) \}$$

$\therefore P(b)$ is true for at least one b in domain
{ some 'b' } is valid



Topic : Existential Generalization

$P(b)$ is true for some b in domain

is valid

$$\exists z (P(z))$$

Equivalences & Implications

$$\textcircled{1} \quad \forall x \{ P(x) \wedge Q(x) \} \quad \begin{array}{c} \xRightarrow{\textcircled{1}} \checkmark \\ \equiv \\ \xleftarrow{\textcircled{2}} \checkmark \end{array} \quad \left\{ \underbrace{\forall x \{ P(x) \}}_T \wedge \underbrace{\forall x \{ Q(x) \}}_T \right\} \equiv T$$

' $\wedge$ ' of P & Q is true for all.

$\therefore P$ & Q both must be true for all

for ' $\wedge$ ' to be true for all ' x ' P should be true and for all ' x ' Q should also be true

$\therefore P \wedge Q$ will also be true for all x

Equivalences & Implications

$$\textcircled{2} \quad \exists x \{ P(x) \vee Q(x) \} \begin{matrix} \xrightarrow{\textcircled{1}} \\ \equiv \\ \xleftarrow{\textcircled{2}} \end{matrix} \{ \exists x \{ P(x) \} \vee \exists x \{ Q(x) \} \}$$

for at least one 'x'
disjunction of P & Q should be true $\Rightarrow$
i.e. at least one of P or Q
must be true for some x

in that case
either $\exists x \{ P(x) \}$ will be true
or $\exists x \{ Q(x) \}$ will be true
or both will be true
So R.H.S. will also be true

Equivalences & Implications

$\therefore$ Not Equivalent

$$\textcircled{3} \quad \underbrace{\forall x \{ P(x) \vee Q(x) \}}_{\text{LHS}} \quad \begin{array}{c} \xrightarrow{\textcircled{1}} \text{invalid} \\ \xleftarrow{\textcircled{2}} \end{array} \quad \underbrace{\{ \forall x \{ P(x) \} \vee \forall x \{ Q(x) \} \}}_{\text{RHS}}$$

For LHS to be true,
disjunction of $P \vee Q$ must
be true for all x .

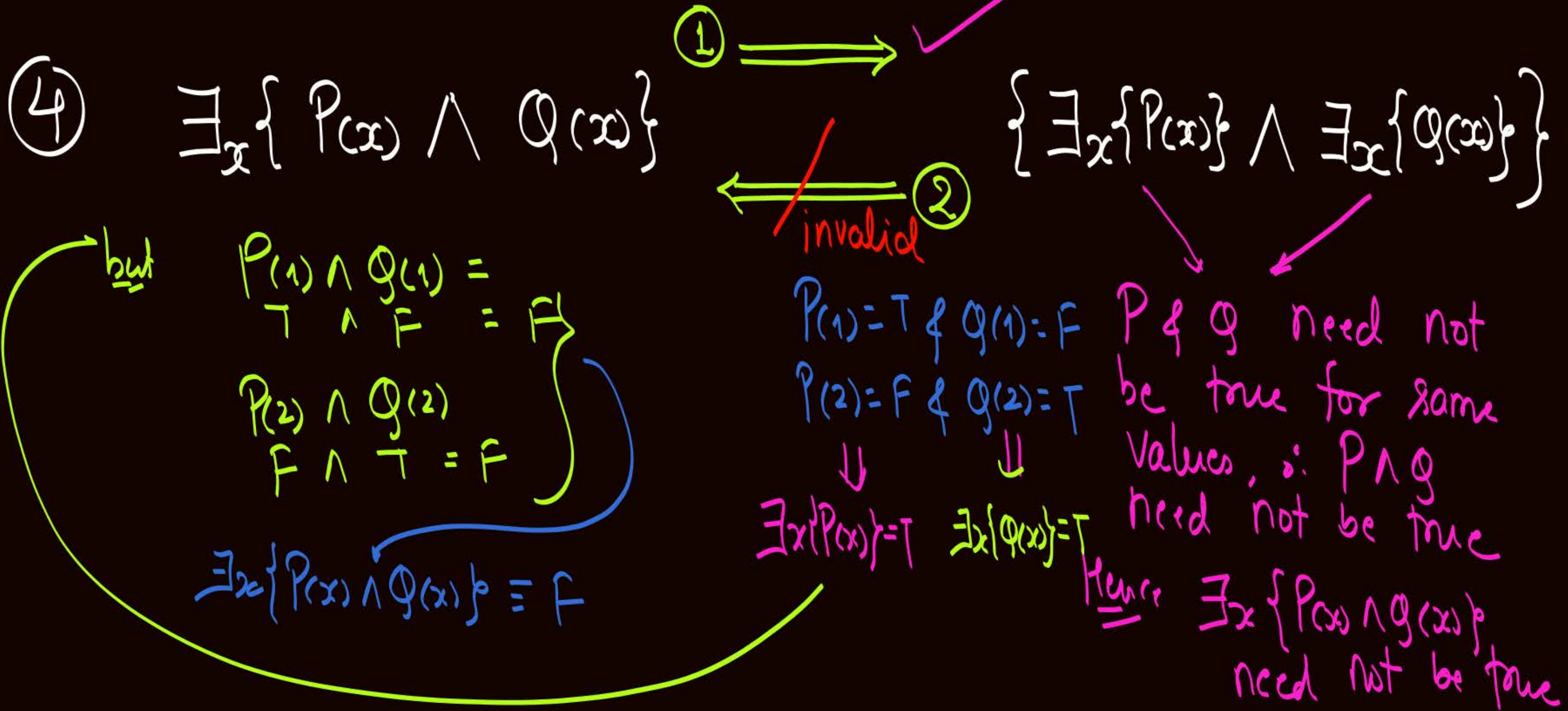
let $U = \{1, 2\}$

$$\begin{aligned} P(1) = T \ \& \ Q(1) = F &\Rightarrow P(1) \vee Q(1) = T \\ P(2) = F \ \& \ Q(2) = T &\Rightarrow P(2) \vee Q(2) = T \end{aligned}$$

$\forall x (P \vee Q)$
is true

$$\begin{aligned} \left. \begin{array}{l} \forall x \{ P(x) \} = F \\ \forall x \{ Q(x) \} = F \end{array} \right\} &\Rightarrow \forall x \{ P(x) \} \vee \forall x \{ Q(x) \} \\ &= F \vee F \\ &= F \\ &\therefore \text{RHS is false} \end{aligned}$$

Equivalences & Implications





Topic : Equivalences & Implications



⑥

$$\forall x [P(x) \leftrightarrow Q(x)] \not\Rightarrow (\forall x [P(x)] \leftrightarrow \forall x [Q(x)])$$

✓ 1. $\forall x [P(x) \wedge Q(x)] \equiv [\forall x P(x)] \wedge [\forall x Q(x)]$

✓ 2. $\exists x [P(x) \vee Q(x)] \equiv [\exists x P(x)] \vee [\exists x Q(x)]$

✓ 3. $[\forall x P(x)] \vee [\forall x Q(x)] \not\Rightarrow \forall x [P(x) \vee Q(x)]$

✓ 4. $\exists x [P(x) \wedge Q(x)] \not\Rightarrow [\exists x P(x)] \wedge [\exists x Q(x)]$

✓ 5. $\forall x [P(x) \rightarrow Q(x)] \not\Rightarrow [\forall x P(x)] \rightarrow [\forall x Q(x)]$



Topic : Some important equivalences

Note:- Here Q is a Predicate formula without any free Occurance of ' x '
 $\Downarrow$
ie, truth value of Q does not vary with ' x '

$\Downarrow$
{ if Q is true then it will be True w.r.t. all x in domain

{ if Q is false, then it will remain false w.r.t. all x in domain

1. $\forall x [P(x) \wedge Q] \equiv \forall x P(x) \wedge Q$

2. $\exists x [P(x) \vee Q] \equiv \exists x P(x) \vee Q$

3. $\forall x [P(x) \vee Q] \equiv \forall x P(x) \vee Q$

4. $\exists x [P(x) \wedge Q] \equiv \exists x P(x) \wedge Q$



Topic : Some important equivalences

1. $\forall x [P(x) \wedge Q] \equiv \forall x P(x) \wedge Q$
 - Case ① When $Q = \text{False}$
if $Q = \text{False}$, then both LHS & R.H.S.
will be false for all values of x
i.e. same truth value of LHS & R.H.S.
 - Case ② When $Q = \text{True}$
(i) For LHS to be true, $P(x)$ must be true
for all x . And in that case $\forall x [P(x)]$ will
also be true, hence RHS will also be true
(ii) For RHS to be true, $\forall x [P(x)]$ must be true
i.e. $P(x)$ must be true for all x , in
that case LHS will also be true
2. $\exists x [P(x) \vee Q] \equiv \exists x P(x) \vee Q$
3. $\forall x [P(x) \vee Q] \equiv \forall x P(x) \vee Q$
4. $\exists x [P(x) \wedge Q] \equiv \exists x P(x) \wedge Q$



Topic : Some important equivalences

1. $\forall x [P(x) \wedge Q] \equiv \forall x P(x) \wedge Q$

2. $\exists x [P(x) \vee Q] \equiv \exists x P(x) \vee Q$

3. $\forall x [P(x) \vee Q] \equiv \forall x P(x) \vee Q$

4. $\exists x [P(x) \wedge Q] \equiv \exists x P(x) \wedge Q$

Similar logic can be analyzed
w.r.t. other three equivalences
as well.



Topic : Some important equivalences

5. $\forall x [P \rightarrow Q(x)] \equiv P \rightarrow \forall x Q(x)$

6. $\exists x [P \rightarrow Q(x)] \equiv P \rightarrow \exists x Q(x)$

Here 'P' is the predicate formula without any free occurrence of 'x'

7. $\forall x [P(x) \rightarrow Q] \equiv \forall x P(x) \rightarrow Q$

8. $\exists x [P(x) \rightarrow Q] \equiv \exists x P(x) \rightarrow Q$

Here, Q is the predicate formula without any free occurrence of x



Topic : Some important equivalences

$$\begin{aligned} 5. \quad & \forall x [P \rightarrow Q(x)] \equiv P \rightarrow \forall x Q(x) \\ & \equiv \forall x \{ \sim P \vee Q(x) \} \\ & \equiv \sim P \vee \forall x \{ Q(x) \} \equiv P \rightarrow \forall x \{ Q(x) \} = \text{RHS} \end{aligned}$$

$$\begin{aligned} 6. \quad & \exists x [P \rightarrow Q(x)] \equiv P \rightarrow \exists x Q(x) \\ & \equiv \exists x \{ \sim P \vee Q(x) \} \\ & \equiv \sim P \vee \exists x \{ Q(x) \} \equiv P \rightarrow \exists x \{ Q(x) \} = \text{RHS} \end{aligned}$$

$$7. \quad \forall x [P(x) \rightarrow Q] \equiv \exists x P(x) \rightarrow Q$$

$$8. \quad \exists x [P(x) \rightarrow Q] \equiv \forall x P(x) \rightarrow Q$$



Topic : Some important equivalences

5. $\forall x [P \rightarrow Q(x)] \equiv P \rightarrow \forall x Q(x)$

6. $\exists x [P \rightarrow Q(x)] \equiv P \rightarrow \exists x Q(x)$

7. $\forall x [P(x) \rightarrow Q] \equiv \exists x P(x) \rightarrow Q$

$$\begin{aligned} \forall x \{(\sim P(x)) \vee Q\} &\equiv \forall x \{ \sim P(x) \} \vee Q \equiv \sim(\exists x \{P(x)\}) \vee Q \\ &\equiv \exists x \{P(x)\} \rightarrow Q \equiv \text{R.H.S.} \end{aligned}$$

8. $\exists x [P(x) \rightarrow Q] \equiv \forall x P(x) \rightarrow Q$

$$\begin{aligned} \equiv \exists x \{ \sim P(x) \vee Q \} &\equiv \exists x \{ \sim P(x) \} \vee Q \equiv \sim(\forall x \{P(x)\}) \vee Q \equiv \forall x \{P(x)\} \rightarrow Q \equiv \text{R.H.S.} \end{aligned}$$

#Q. Let $P(x)$ and $Q(x)$ be arbitrary predicates. Which of the following statements is always TRUE?

☒ **A** $(\forall x(P(x) \vee Q(x))) \Rightarrow ((\forall x P(x)) \vee (\forall x Q(x)))$

Handwritten: Green double arrow and checkmark above the implication symbol.

☒ **B** $(\forall x(P(x) \Rightarrow Q(x))) \Rightarrow ((\forall x P(x)) \Rightarrow (\forall x Q(x)))$

Handwritten: Red double arrow above the implication symbol.

☒ **C** $(\forall x(P(x) \Rightarrow \forall x Q(x))) \Rightarrow (\forall x (P(x) \Rightarrow Q(x)))$

Handwritten: Green double arrow and checkmark above the implication symbol.

☒ **D** $((\forall x(P(x)) \Leftrightarrow (\forall x Q(x))) \Rightarrow (\forall x(P(x) \Leftrightarrow Q(x))))$

Handwritten: Green double arrow and checkmark above the implication symbol.

#Q. Which of the following predicate calculus statements is/are valid Let $U = \{1, 2\}$

☒ **A**

$$(\forall x)P(x) \vee (\forall x)Q(x) \rightarrow (\forall x)\{P(x) \vee Q(x)\}$$

☒ **B**

$$(\exists x)P(x) \wedge (\exists x)Q(x) \rightarrow (\exists x)\{P(x) \wedge Q(x)\}$$

☒ **C**

$$(\exists x)\{P(x) \vee Q(x)\} \rightarrow (\forall x)P(x) \vee (\forall x)Q(x)$$

☒ **D**

$$(\exists x)\{P(x) \vee Q(x)\} \rightarrow \{\exists x P(x) \vee \forall x Q(x)\}$$

$\equiv$

$\exists x$

let $P = \text{False for all } x$
 $\& Q(1) = T \& Q(2) = F$

$\therefore P(1) \vee Q(1) = T$
 $\quad \quad \quad F \quad \quad T$
 $\&$
 $P(2) \vee Q(2) = F$
 $\quad \quad \quad F \quad \quad F$

$\therefore \exists x\{P(x) \vee Q(x)\}$
 is true
 So LHS
 is true

but $P = \text{False for all } x \therefore \exists x\{P(x)\} = F$
 $Q(2) = F \therefore \forall x\{Q(x)\} = F$
 Hence $\exists x\{P(x)\} \vee \forall x\{Q(x)\} = F \vee F = F$

H.W.
#Q. Which of the following predicate calculus statements is/are valid

A $\forall(x)(P(x) \Rightarrow Q(x)) \Rightarrow ((\forall(x)P(x)) \Rightarrow (\forall(x)Q(x)))$

B $\exists x(P(x) \vee Q(x)) \Rightarrow ((\exists xP(x)) \Rightarrow (\exists xQ(x)))$

C $\exists x(P(x) \wedge Q(x)) \Leftrightarrow ((\exists xP(x)) \wedge (\exists xQ(x)))$

D $\forall(x)\exists y P(x, y) \Rightarrow \exists y \forall(x)P(x, y)$

H.W.

#Q. Which of the following is a valid first order formula? (Here α and β are first order formulae with x as their only free variable)

~~A~~ $((\forall x)[\alpha] \Rightarrow (\forall x)[\beta]) \Rightarrow (\forall x)[\alpha \Rightarrow \beta]$
 $\neq (\forall x \{P(x)\} \rightarrow \forall x \{Q(x)\}) \rightarrow \forall x \{P(x) \rightarrow Q(x)\}$

B $((\forall x)[\alpha] \Rightarrow (\exists x)[\alpha \wedge \beta])$

C $((\forall x)[\alpha \vee \beta] \Rightarrow (\exists x)[\alpha] \Rightarrow (\forall x)[\alpha])$

D $((\forall x)[\alpha \Rightarrow \beta] \Rightarrow ((\forall x)[\alpha]) \Rightarrow (\forall x)[\beta])$

[MSQ]

#Q. Which of the following first order formulae is logically valid? Here $\alpha(x)$ is a first-order formula with x as a free variable and β is a first-order formula with no free variable.

- A** $[\beta \rightarrow (\exists x, \alpha(x))] \rightarrow [\forall x, \beta \rightarrow \alpha(x)]$
- B** $[\exists x, \beta \rightarrow (\alpha(x))] \rightarrow [\beta \rightarrow (\forall x, \alpha(x))]$
- C** $[(\exists x, \alpha(x)) \rightarrow \beta] \rightarrow [\forall x, \alpha(x) \rightarrow \beta]$
- D** $[(\forall x, \alpha(x)) \rightarrow \beta] \rightarrow [\forall x, \alpha(x) \rightarrow \beta]$

H.W.

#Q. Which one of the following predicate formulae is NOT logically valid? Note that W is a predicate formula without any free occurrence of x .

A $\exists x(p(x) \wedge W) \equiv (\exists x p(x)) \wedge W$

B $\forall x(p(x) \rightarrow W) \equiv (\forall x p(x)) \rightarrow W$

C $\exists x(p(x) \rightarrow W) \equiv (\forall x p(x)) \rightarrow W$

D $\forall x(p(x) \vee W) \equiv (\forall x p(x)) \vee W$

HW
#Q. What is the correct translation of the following statement into mathematical logic?

“some real numbers are rational”

A $\exists x (real(x) \vee rational(x))$

B $\exists x (real(x) \wedge rational(x))$

C $\forall x (real(x) \rightarrow rational(x))$

D $\forall x (rational(x) \rightarrow real(x))$

1-2-21
[MSQ]

#Q. What is the logical translation of the following statement?

"None of my friends are perfect"

- A** $\exists x (F(x) \wedge \sim P(x))$
- B** $\exists x (\sim F(x) \wedge \sim P(x))$
- C** $\exists x (\sim F(x) \wedge P(x))$
- D** $\sim (\exists x (F(x) \wedge P(x)))$

[MSQ]

#Q. The CORRECT formula of the sentence “not all rainy days are cold”

- A** $\forall d (rainy(d) \wedge \sim cold(d))$
- B** $\exists d (\sim rainy(d) \rightarrow cold(d))$
- C** $\forall d (\sim rainy(d) \rightarrow cold(d))$
- D** $\exists d (rainy(d) \wedge \sim cold(d))$

H.W. #Q. Which one the following is the most appropriate logical formula to represent the statement “gold and silver ornaments are precious”. The following notations are used:

$G(x)$: x is a gold ornament

$S(x)$: x is a silver ornament

$P(x)$: x is precious

A

$$\forall x (P(x) \rightarrow G(x) \wedge S(x))$$

B

$$\exists x ((G(x) \wedge S(x)) \rightarrow P(x))$$

C

$$\forall x ((G(x) \wedge S(x)) \rightarrow P(x))$$

D

$$\forall x ((G(x) \vee S(x)) \rightarrow P(x))$$

[MSQ]

#Q. Which one of the first order predicate calculus statements given below correctly expresses the following English statements? "Tigers and lions attack if they are hungry or threatened".

- A** $\forall x[(tiger(x) \wedge lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \rightarrow attacks(x))]$
- B** $\forall x[(tiger(x) \vee lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \wedge attacks(x))]$
- C** $\forall x[(tiger(x) \wedge lion(x)) \rightarrow (attacks(x) \rightarrow (hungry(x) \vee threatened(x)))]$
- D** $\forall x[(tiger(x) \vee lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \rightarrow attacks(x))]$



2 mins Summary



Topic

Sentences with multiple quantifiers

Topic

Relationship diagram

→ Rules of inference w.r.t. Predicate logic

→ Equivalences & Implications.

→ Practice questions

THANK - YOU